8

Irradiation

Ionising radiation takes the form of γ-rays from isotopes or, commercially to a lesser extent, from X-rays and electrons. It is permitted in 38 countries to preserve foods by destruction of micro-organisms or inhibition of biochemical changes (Table 8.1). Applications and details of the technology have been comprehensively reviewed by Wilkinson and Gould (1996).

The main advantages of irradiation are as follows:

- there is little or no heating of the food and therefore negligible change to sensory characteristics
- packaged and frozen foods may be treated
- fresh foods may be preserved in a single operation, and without the use of chemical preservatives
- energy requirements are very low
- changes in nutritional value of foods are comparable with other methods of food preservation
- processing is automatically controlled and has low operating costs.

A major disadvantage is the high capital cost of irradiation plant, but concern over the use of food irradiation has also been expressed by some (for example Webb and Lang (1990) and Webb and Henderson (1986)). Other concerns, for example over operator safety, are discussed by Webb and Lang (1987) and by Welt (1985). They describe the main problems as:

- the process could be used to eliminate high bacterial loads to make otherwise unacceptable foods saleable
- if spoilage micro-organisms are destroyed but pathogenic bacteria are not, consumers will have no indication of the unwholesomeness of a food
- there will be a health hazard if toxin-producing bacteria are destroyed after they have contaminated the food with toxins
- the possible development of resistance to radiation in micro-organisms
- loss of nutritional value

- until recently, inadequate analytical procedures for detecting whether foods have been irradiated
- public resistance due to fears of induced radioactivity or other reasons connected to concerns over the nuclear industry.

These concerns have been addressed by the Joint FAO/IAEA/WHO Expert Committee on the Wholesomeness of Irradiated Food (JECFI) who concluded that the maximum average dose of 10 kGy 'presents no toxicological hazard and no special nutritional or microbiological problems in foods' (World Health Organisation, 1977, 1981). This was supported by the Advisory Committee on Irradiated and Novel Foodstuffs (Anon., 1986). The JECFI recommendations were then formed into a standard by the Codex Alimentarius Commission who recommended an international code for the operation of radiation facilities. In most countries where irradiation is permitted, labelling regulations are in force which place a requirement on manufacturers to indicate that the food or any listed ingredients have been treated by irradiation. Additionally, wholesale foods are required to be labelled with the phrase 'Treated by irradiation, do not irradiate again'. This, together with a programme of research into methods to detect whether foods have been irradiated, are in part intended to change negative public perceptions of irradiated foods.

Table 8.1 Applications of food irradiation

Application	Dose range (kGy)	Examples of foods	Countries with commercial processing
Sterilisation	7–10	Herbs, spices	Belgium, Canada, Croatia, Czech Republic, Denmark, Finland, Israel, Korea (Rep.), Mexico, South Africa, USA, Vietnam
	Up to 50	Long term ambient storage of meat (outside permitted dose)	None
Sterilisation of packaging materials	10–25	Wine corks	Hungary
Destruction of pathogens	2.5–10	Spices, frozen poultry, meat, shrimps	Belgium, Canada, Croatia, Czech Republic, Denmark, Finland, France, Iran, Netherlands, South Africa, Thailand, Vietnam
Control of moulds	2–5	Extended storage of fresh fruit	China, South Africa, USA
Extension of chill life from 5 days to 1 month	2–5	Soft fruit, fresh fish and meat at 0–4ºC	China, France, Netherlands, South Africa, USA
Inactivation/control of parasites	0.1–6	Pork	–
Disinfestation	0.1–2	Fruit, grain, flour, cocoa beans, dry foods	Argentina, Brazil, Chile, China
Inhibition of sprouting	0.1–0.2	Potatoes, garlic, onions	Algeria, Bangladesh, China, Cuba

Adapted from Ley (1987), Guise (1986a), Goresline (1982), Anon. (1985), Loaharanu (1995) and Wilkinson and Gould (1996).

8.1 Theory

The units used are given as a footnote to this page.[1] Details of the physical and chemical processes involved in the decay of radioactive materials to produce α-, β- and γ radiation, X-rays and free electrons are described by Desrosier and Rosenstock (1960).

γ-rays and electrons are distinguished from other forms of radiation by their ionising ability (that is they are able to break chemical bonds when absorbed by materials). The products of ionisation may be electrically charged (ions) or neutral (free radicals). These then further react to cause changes in an irradiated material known as radiolysis. It is these reactions that cause the destruction of micro-organisms, insects and parasites during food irradiation.

In foods that have a high moisture content, water is ionised by radiation. Electrons are expelled from water molecules and break the chemical bonds. The products then recombine to form hydrogen, hydrogen peroxide, hydrogen radicals (H·), hydroxyl radicals (OH·) and hydroperoxyl radicals (HO_2·) (Fig. 8.1).

The radicals are extremely short lived (less than 10^{-5} s) but are sufficient to destroy bacterial cells. Similar radicals are also present in non-irradiated foods owing to:

- the action of enzymes (for example lipoxygenases and peroxidases)
- the oxidation of fats and fatty acids
- the degradation of fat soluble vitamins and pigments.

In addition, reactive oxygen and its derivatives are produced in foods by peroxidases, xanthine oxidase and amino acid oxidase. Fat-soluble components and essential fatty acids are therefore lost during irradiation and some foods (for example dairy products) are unsuitable for irradiation owing to the development of rancid off-flavours. The presence of oxygen accelerates this process and meat is therefore irradiated in vacuum packs.

(a)

$$H_2O \rightarrow H_2O^+ + e^-$$
$$e^- + H_2O \rightarrow H_2O^-$$
$$H_2O^+ \rightarrow H^+ + OH\cdot$$
$$H_2O^- \rightarrow H\cdot + OH^-$$

(b)

$$H\cdot + H\cdot \rightarrow H_2$$
$$\text{or } OH\cdot + OH\cdot \rightarrow H_2O_2$$
$$\text{or } H\cdot + OH\cdot \rightarrow H_2O$$
$$\text{or } H\cdot + H_2O \rightarrow H_2 + OH\cdot$$
$$\text{or } OH\cdot + H_2O_2 \rightarrow H_2O + HO_2\cdot$$
$$H\cdot \rightarrow + O_2HO_2.$$

Fig. 8.1 (a) Ionisation of water (after Robinson (1986)); (b) formation of free radicals during irradiation (after Hughes (1982)).

1. *Summary of units*:

Becqerel (Bq):	One unit of disintegration per second. Curie (Ci) = 3.7×10^{10} Bq.
Half-life:	The time taken for the radioactivity of a sample to fall to half its initial value.
Electron volt (eV):	Energy of radiation (usually as mega-electron volts (MeV)). 1 eV = 1.602×10^{-19} J.
Grays (Gy):	Absorbed dose (where 1 Gy is the absorption of 1 J of energy per kilogram of food). Previously rads (radiological units) were used where 1 rad = 10^{-2} J kg^{-1}. 1 Gy therefore equals 100 rads.

8.2 Equipment

Irradiation equipment consists of a high-energy isotope source to produce γ-rays or, less commonly, a machine source to produce a high-energy electron beam. γ-radiation from cobalt-60 (^{60}Co) or caesium-137 (^{137}Cs) is used in most commercial plants. ^{60}Co emits γ-rays at two wavelengths which have energies of 1.17 MeV and 1.33 MeV respectively. The activity of the ^{60}Co or ^{137}Cs sources is rated at (222–370) $\times 10^{10}$ Bq g^{-1} (or 10^{13} Bq kg^{-1}). This generates 15 kW/M Ci (15 kW per/3.7 $\times 10^{16}$ Bq). The residence time of the food is determined by the dose required (Table 8.1) and the power output of the source.

An isotope source cannot be switched off and so is shielded within a pool of water below the process area, to allow personnel to enter. In operation the source is raised, and packaged food is loaded onto automatic conveyors and transported through the radiation field in a circular path. This makes maximum use of the emitted radiation and ensures a uniform dose. Isotope sources require a more complex materials-handling system than that used with machine sources (Fig. 8.2).

Machine sources are electron accelerators which consist of a heated cathode to supply electrons and an evacuated tube in which electrons are accelerated by a high-voltage electrostatic field. Either the electrons are used directly on the food, or a suitable target material is bombarded to produce X-rays. The main advantages of machine sources are:

- they can be switched off
- the electron beams can be directed over the packaged food to ensure an even dose distribution.

Handling equipment is therefore relatively simple. However, machine sources are expensive and relatively inefficient in producing radiation.

Radiation is contained within the processing area by the use of thick concrete walls and lead shielding. Openings in the shielding, for entry of products or personnel, must be carefully constructed to prevent leakage of radiation. A dose of 5 Gy is sufficient to kill an operator and it is therefore essential that even at the lowest commercial doses (0.1 kGy), stringent safety procedures are in place to prevent the source from being raised when personnel are present and to prevent entry to the building during processing.

^{60}Co has a half-life of 5.26 years and therefore requires the replacement of 12.3% of the activity each year to retain the rated output of the plant. Continuous-processing is

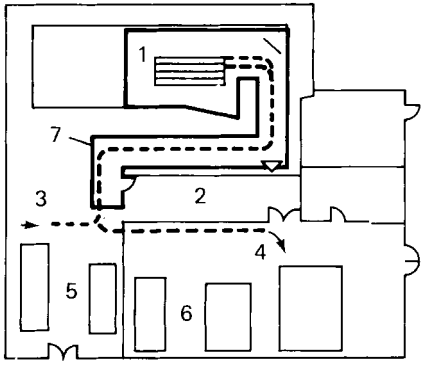


Fig. 8.2 Isotope irradiation plant: 1, irradiation chamber; 2, control room; 3, infeed conveyor; 4, outlet conveyor; 5, raw food store; 6, irradiated product store; 7, concrete shielding wall. (After Farrall (1976).)

Table 8.2 Energy used in irradiation of chicken pieces compared to some other processes

Process	Energy use (kJ/kg)
Refrigerated (few days shelf life)	17 760
Frozen and stored for 3–5 weeks	46 600
Canned	20 180
Irradiated and refrigerated (few weeks shelf life)	17 860

Adapted from Brynjolfsson (1978).

therefore desirable for economic operation of plant with a continuously decaying source (Wilkinson, 1986). The energy savings incurred when irradiation/refrigeration of chicken pieces is compared to other processes are described in Table 8.2.

8.2.1 Measurement of radiation dose

Dosimeters are made from a number of materials, including photographic film, Perspex and cobalt glass, which are described by McLaughlin *et al.* (1982). Polyvinylchloride (PVC) dosimeters are impregnated with a dye. Hydrogen chloride is released from the PVC by irradiation and it produces a qualitative or quantitative change in the colour of the dye to indicate the dose received.

8.2.2 Dose distribution

Penetration of γ-radiation depends on the density of the food as well as the energy of the rays. Because radiation is absorbed as it passes through the food, the outer parts receive a higher dose than do inner parts. At a density of 1000 kg m^{-3}, half of the rays are absorbed in 11 cm. Halving the density approximately doubles the depth of penetration. For a given food there is thus a limit on both the maximum dose permitted at the outer edge (D_{max}), due to either unacceptable organoleptic changes or legislation, and a minimum limit (D_{min}) to achieve the desired effects of microbial destruction. The uniformity of dose distribution can be expressed as a ratio of D_{max}:D_{min}. For foods that are sensitive to radiation, such as chicken, this ratio should be as low as possible and not more than about 1.5. Other foods, for example onions, can tolerate a ratio of around 3 without unacceptable changes. The distribution of dose can be controlled by adjusting the thickness of the packed product and irradiating from both sides.

Dosimeters are placed at points throughout the package to determine the dose received and to ensure that the D_{max}:D_{min} ratio is achieved.

High energy electrons have a lower penetration than γ-rays and are not suitable for bulk foods. They are used for thin packages or for surface treatments. The selection of a radiation source therefore depends on the type of product and its density, the dimensions of the package and the reason for the treatment.

8.3 Effect on micro-organisms

The reactive ions produced by irradiating foods (Fig. 8.1) injure or destroy micro-organisms immediately, by changing the structure of cell membranes and affecting metabolic enzyme activity. However, a more important effect is on deoxyribonucleic acid (DNA) and ribonucleic acid molecules in cell nuclei, which are required for growth and replication. The

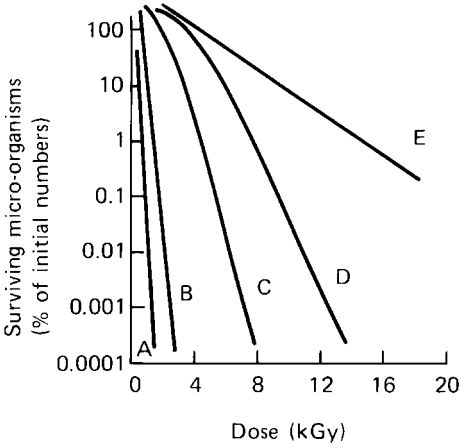


Fig. 8.3 Microbial destruction by irradiation: A, *Pseudomonas* sp.; B, *Salmonella* sp.; C, *Bacillus cereus*; D, *Deinococcus radiodurans*; E, typical virus.
(After Gould (1986).)

effects of irradiation only become apparent after a period of time, when the DNA double-helix fails to unwind and the micro-organism cannot reproduce by cell division.

The rate of destruction of individual cells depends on the rate at which ions are produced and inter-react with the DNA, whereas the reduction in cell numbers depends on the total dose of radiation received. The sensitivity of micro-organisms to radiation is expressed as the D_{10} value (the dose of radiation that reduces the microbial population to 10% of its initial value) by analogy with thermal destruction (Chapter 1). Theoretically, a logarithmic reduction in microbial numbers with increasing dose is expected. As in other food preservation methods the rate of destruction varies with microbial species, and some bacterial species contain more than one molecule of DNA whereas others are capable of repairing damaged DNA. The rate of destruction is not therefore linear with received dose (Fig. 8.3).

A simple guide is that, the smaller and simpler the organism, the higher the dose of radiation that is needed to destroy it. Viruses are very resistant to irradiation and are unlikely to be affected by the dose levels used in commercial processing. Spore forming species (for example *Clostridium botulinum* and *Bacillus cereus*), and those that are able to repair damaged DNA rapidly (for example *Deinococcus radiodurans*) are more resistant than vegetative cells and non-spore forming bacteria. Insects and parasites such as tape worms or trichinella require lower doses (e.g. *Toxoplasma gondii* is inactivated at doses of 0.25 kGy and *Trichinella spiralis* at 0.3 kGy (Olson, 1998)). Examples of *D*-values for important pathogens are given in Table 8.3.

8.4 Applications

The radiation dose administered to a food depends on the resistance of the organisms present and the objective of the treatment. The maximum recommended dose for foods is 15 kGy, with the average dose not exceeding 10 kGy (World Health Organisation, 1977, 1981, 1994). The types of irradiation process can be categorised by the intention of processing and the dose used.

Table 8.3 *D*-values of important pathogens

Pathogen	*D*-value (kGy)	Irradiation temperature (ºC)	Suspending medium
A.hydrophilia	0.14–0.19	2	Beef
C. jejuni	0.18	2–4	Beef
Escherichia coli 0157:H7	0.24	2–4	Beef
L. monocytogenes	0.45	2–4	Chicken
Salmonella sp.	0.38–0.77	2	Chicken
Staphlococcus aureus	0.36	0	Chicken
Yersinia enterocolitica	0.11	25	Beef
Clostridium botulinum (spores)	3.56	−30	Chicken

Adapted from Olson (1998).

8.4.1 Sterilisation (or 'radappertisation')

Although it is technically possible to sterilise meats and other products, the dose required exceeds the current limit of 10 kGy. A dose of 48 kGy is needed for a 12*D* reduction of *Cl. botulinum* (Lewis, 1990) but such a high dose would make products such as meat organoleptically unacceptable. There is thus little commercial interest in sterilisation, with the exception of herbs and spices which are frequently contaminated by heat resistant, spore-forming bacteria. These products can be sterilised using a dose of 8–10 kGy (Table 8.1), which reduces the microbial load to an acceptable level without significant loss of volatile oils, the main quality characteristic. The main advantage of irradiating spices is the replacement of chemical sterilisation using ethylene oxide, which has been banned by the European Union since 1991 as a result of concerns over residues in the product and safety of workers handling the gas.

8.4.2 Reduction of pathogens (or 'radicidation')

Food poisoning bacteria (for example *Salmonella typhimurium*) are less resistant to radiation than for example *Clostridium botulinum*, and doses of 3–10 kGy are sufficient for destruction (Guise, 1986a). This is likely to become one of the most important applications of food irradiation as the incidence of food poisoning is steadily increasing in many countries (Guise, 1986b; Loaharanu, 1995). Fresh poultry carcasses irradiated with a dose of 2.5 kGy are virtually free of *Salmonella* spp. and the shelf life is doubled when the product is held below 5ºC. Higher doses, up to 10 kGy, can be applied to frozen poultry or shellfish (−18ºC) to destroy *Campylobacter* spp. *Escherichia coli* 0157:H7 or *Vibrio* sp. (e.g. *V. cholerae, V. parahaemolyticus, V. vulnificus*) without causing the unacceptable organoleptic changes that occur in fresh meat. These applications have been reviewed by Crawford and Ruff (1996).

8.4.3 Prolonging shelf life (or 'radurisation')

Relatively low doses (Table 8.1) are needed to destroy yeasts, moulds and non-spore forming bacteria. This process is used to increase shelf life by an overall reduction in vegetative cells. Bacteria which survive irradiation are more susceptible to heat treatment and the combination of irradiation with heating is therefore beneficial in causing a greater reduction in microbial numbers than would be achieved by either treatment alone (Gould, 1986).

Two potential problems of irradiation that are raised with respect to microbial destruction are:

1. that by destroying spoilage micro-organisms and not destroying pathogenic bacteria a valuable indicator of unwholesomeness is removed
2. that the destruction of toxin-producing bacteria after they have contaminated the food with toxins is a health hazard.

These are problems associated with poor manufacturing practice and occur in other types of food processing (for example aflatoxin production by *Aspergillus* species in nuts and grains). It is therefore essential that codes of Good Manufacturing Practice (for example Anon., 1998) are adhered to.

8.4.4 Control of ripening

Some types of fruits and vegetables, such as strawberries and tomatoes, can be irradiated to extend their shelf life about 2–3 times when stored at 10ºC. A combination of irradiation and modified atmosphere packaging (Chapter 20) has been shown to have a synergistic effect and as a result, a lower radiation dose can be used to achieve the same effect. However, these products should be ripe before irradiation as the ripening process is inhibited. Ripening and maturation of fruits and vegetables are arrested by inhibiting hormone production and interrupting the biochemical processes of cell division and growth. It should be noted, however, that enzymic spoilage of foods is not entirely prevented by irradiation and a separate heat treatment is required for prolonged storage. A two-fold increase in shelf life of mushrooms and inhibition of cap opening can be achieved by doses of 2–3 kGy.

8.4.5 Disinfestation

Grains and tropical fruits may be infested with insects and larvae, which lowers their export potential and requires a quarantine period for disinfestation. Low doses, below 1 kGy, are effective for disinfestation and also extend the shelf life by delaying ripening. Further advantages are the avoidance of pesticides such as ethylene dibromide and methyl bromide which are now banned or restricted in many countries (Loaharanu, 1995) as a result of concern over toxic residues in foods or, in the case of methyl bromide, because of its potential to deplete the ozone layer.

8.4.6 Inhibition of sprouting

The technology is effective in inhibiting sprouting of potatoes and in Japan for example, doses of about 150 Gy have been used since 1973 on potatoes intended for further processing (Stevenson, 1990). Similar doses are effective in preventing sprouting of onions and garlic.

8.5 Effect on foods

8.5.1 Induced radioactivity

At recommended doses, ^{60}Co and ^{137}Cs have insufficient emission energies to induce radioactivity in the food. Machine sources of electrons and X-rays do have sufficient

energy, but the levels of induced radioactivity are insignificant (2% of the acceptable radiation dose in the worst case and 0.0001% under realistic processing and storage conditions) (Gaunt, 1986).

8.5.2 Radiolytic products

The ions and radicals produced during irradiation (Section 8.1) are capable of reacting with components of the food to produce radiolytic products. The extent of radiolysis depends on the type of food and the radiation dose employed. However, the majority of the evidence from feeding experiments, in which animals were fed irradiated foods and high doses of radiolytic products, indicates that there are no adverse effects. Radiolytic type products can also arise from other methods of food processing (Gaunt, 1986) and as a result it has, until recently, been difficult to devise a test to determine whether foods have been irradiated.

8.5.3 Nutritional and sensory value

At commercial dose levels, ionising radiation has little or no effect on the digestibility of proteins or the composition of essential amino acids (Josephson *et al.*, 1975). At higher dose levels, cleavage of the sulphydryl group from sulphur amino acids in proteins causes changes in the aroma and taste of foods. Carbohydrates are hydrolysed and oxidised to simpler compounds and, depending on the dose received, may become depolymerised and more susceptible to enzymic hydrolysis. However, there is no change in the degree of utilisation of the carbohydrate and hence no reduction in nutritional value. The effect on lipids is similar to that of autoxidation, to produce hydroperoxides and the resulting unacceptable changes to flavour and odour. The effect is reduced by irradiating foods while frozen, but foods that have high concentrations of lipid are generally unsuitable for irradiation.

There is conflicting evidence regarding the effect on vitamins as many studies have used vitamin solutions, which show greater losses than those found in the heterogeneous mixtures of compounds in foods. Water soluble vitamins vary in their sensitivity to irradiation. The extent of vitamin loss also depends on the dose received and the type and physical state of food under investigation. There is also conflicting evidence at low dose levels. For example, in grain disinfestation there are little or no vitamin losses, whereas inhibition of sprouting is variously reported to cause 0% and 28% loss of vitamin C at 0.1 kGy and 0.11 kGy respectively (Faizur Rahman, 1975; Gounelle *et al.*, 1970). The effects on thiamine in meat and poultry are likewise inconsistent but other vitamins of the B group are largely unaffected (Table 8.4). The order of sensitivity is reported as thiamin > ascorbic acid > pyridoxine > riboflavin > folic acid > cobalamin > nicotinic acid. Fat-soluble vitamins vary in their susceptibility to radiation. Vitamins D and K are largely unaffected whereas vitamins A and E undergo some losses, which vary according to the type of food examined. The order of sensitivity is vitamin E > carotene > vitamin A > vitamin K > vitamin D (World Health Organisation, 1994). A comparison of irradiated and heat sterilised chicken meat (Table 8.5) indicates similar levels of vitamin loss.

In summary, the consensus of opinion is that, at commercial dose levels, irradiation causes no greater damage to nutritional quality than other preservation operations used in food processing. Changes in nutritional quality are described in detail by Kraybill (1982).

Table 8.4 Effect of irradiation on water-soluble vitamins in selected foods

Food	Treatment (kGy)	Loss (%)					
		Thiamin	Riboflavin	Niavin	Pyridoxine	Pantothenic acid	Vitamin B_{12}
Beef	4.7–7.1	60	4	14	10	–	–
Pork	4.5	15	22	22	2	–	–
Haddock	1.5	22	0	0	+15	+78	10
Wheat	2.0	12	13	9	–	–	–
Flour	0.3–0.5	0	0	11	0	–	–

+ apparent increase. Adapted from Brooke *et al*. (1966) and Josephson *et al*. (1975).

Table 8.5 A comparison of vitamin contents of heat sterilised and irradiated (58 kGy at 25ºC) chicken meat

Vitamin	Vitamin concentration (mg/kg dry weight)[a]			
	Frozen control	Heat sterilised	γ-irradiated	Electron irradiated
Thiamin HCl	2.31	1.53[b]	1.57[b]	1.98
Riboflavin	4.32	4.60	4.46	4.90[c]
Pyridoxine	7.26	7.62	5.32	6.70
Nicotinic acid	212.9	213.9	197.9	208.2
Pantothenic acid	24.0	21.8	23.5	24.9
Biotin	0.093	0.097	0.098	0.013
Folic acid	0.83	1.22	1.26	1.47[c]
Vitamin A	2716	2340	2270	2270
Vitamin D	375.1	342.8	354.0	466.1
Vitamin K	1.29	1.01	0.81	0.85
Vitamin B_{12}	0.008	0.016[c]	0.014[c]	0.009

[a] Concentrations of vitamin D and vitamin K are given as IU/kg.
[b] Significantly lower than frozen control.
[c] Significantly higher than frozen control.
From Satin (1993).

8.6 Effect on packaging

Radiation is able to penetrate packaging materials and therefore reduces the risk of post-processing contamination and allows easier handling of products. However, packaging materials are themselves subject to changes induced by radiation (Table 8.6) and may produce low molecular weight hydrocarbons and halogenated polymers, which have the potential to migrate into the product and produce unacceptable tainting. Careful choice of packaging materials as well as adhesives, additives and printing materials is necessary to prevent contamination of the food with radiolytic products.

8.7 Detection of irradiated foods

There has been considerable research since the late 1980s to develop and validate a series of detection methods that can be used by enforcement officers to detect whether a food has been irradiated. A review of the international collaboration to develop standard

Table 8.6 Changes to packaging materials caused by irradiation

Packaging material	Maximum dose (kGy)	Effect of radiation above maximum dose
Polystyrene	5000	–
Polyethylene	1000	–
PVC	100	Browning, evolution of hydrogen chloride
Paper and board	100	Loss of mechanical strength
Polypropylene	25	Becomes brittle
Glass	10	Browning

Adapted from the data of Guise (1986a) and McLaughlin *et al*. (1982).

methods is given by Delincée (1998). Because irradiation produces no major chemical, physical or sensory changes to foods at commercial doses, detection methods focus on minute changes in chemical composition, physical or biological changes to the food. These methods are based on either detecting products formed by irradiation, physical changes such as cell membrane damage or determining the ratio of live:dead bacteria. At present there are a number of techniques under development, but it is not likely that one method will be applicable to all irradiated foods. These developments have been reviewed by Delincée (1998). So far the following methods have either been validated or are likely to prove useful.

8.7.1 Physical methods

Electron spin resonance spectroscopy

Electron spin resonance (ESR) spectroscopy detects radicals produced by irradiation, which are stable in solid or dry components of foods (e.g. bones). It has been used to detect irradiated bone-containing meat and fish products and shellfish, and may have application to a wide range of food products, including foods that contain crystalline cellulose and have a low moisture content (e.g. pistachio nuts and paprika). It is a non-destructive method that is specific, rapid and relatively simple, although the high cost of ESR spectrometers may limit its application.

Thermoluminescence

Thermoluminescence (TL) is based on the emission of light, when energy trapped in crystalline lattices during irradiation is released by heating the food. These lattices may be contaminating dust (silicate materials) on spices, minerals from seabeds in the intestines of shellfish, or minerals in fruits and vegetables. The minerals are isolated from the food and when they are heated in a controlled way, the stored energy is released as light and measured by a sensitive photon counter. The method is widely applicable and gives an unequivocal determination that a food has been irradiated, but it has a number of disadvantages: it is laborious because a sufficient amount of minerals (a few mg), free of organic matter, has to be prepared; strict procedures are needed to prevent contamination by dust in the laboratory; TL analysers are expensive; and a radiation source is needed for standardisation of the method. A recent adaptation of the method is photostimulated luminescence (PSL) in which pulsed infrared light is used instead of heat to release the stored energy. The method does not require isolation of the minerals and a small sample of food can be used directly to obtain a result within a few minutes. It is also more efficient than TL in detecting an irradiated component in a mixture of materials and may be capable of being developed into a ‘spot-test’ for irradiation.

Other methods that detect physical changes are measurement of electrical impedance, changes in viscosity, electric potential, nuclear magnetic resonance and near-infrared spectroscopy (World Health Organisation, 1994).

8.7.2 Chemical methods

The lack of chemical changes to foods as a result of irradiation has caused difficulties in devising suitable tests. One method that has been developed is analysis of radiolytic hydrocarbons produced from fats using gas chromatography (GC). Although the products are formed in other types of food processing, they have a characteristic distribution pattern following irradiation. The use of liquid chromatography coupled to GC increases the sensitivity of the method (Delincée, 1998).

2-alkylcyclobutanones

2-alkylcyclobutanones (2-CBs) are radiolytic products formed from fatty acids, but are not found as a result of other degradative processes. They are detected using mass spectrometry after separation by gas chromatography and can be identified as positive markers to indicate that a food has been irradiated (Crone *et al*., 1993; Stevenson, 1994). The method has been used to detect irradiated poultry meat, pork, beef or lamb meat and liquid whole egg. More recently the test has been used to detect irradiated exotic fruits, including mango, papaya and avocado (Stewart *et al.*, 1998). Other methods that have been used to detect 2-CBs include thin-layer chromatography (TLC), high pressure liquid chromatography (HPLC) and supercritical fluid extraction/TLC.

Detection of DNA fragmentation by microgel electrophoresis of single cells or nuclei has shown good results but the test is limited to foods that have not been heat treated, as this also results in DNA fragmentation. There are also prospects for routine use of enzyme linked immunosorbant assay (ELISA) tests as simpler and more rapid methods to detect 2-CBs and dihydrothymidine, a DNA breakdown product of irradiation. Details are given by Nolan *et al*. (1998) and Tyreman *et al.* (1998) respectively. In each case the monoclonal antibody used in the assay is raised against the particular marker chemical and subjected to the ELISA testing procedure. The results give both qualitative and quantitative identification of the marker in irradiated samples of food and may be developed into a positive ‘spot-test’ for irradiation.

8.7.3 Biological methods

The Limulus Amoebocyte Lysate test in conjunction with a Gram-negative bacteria count (LAL/GNB), is used to estimate the reduced viability of micro-organisms in a food after irradiation, measuring both dead and alive micro-organisms. This method and the Direct Epifluorescent Filter Technique with an aerobic plate count (DEFT/APC) give information about the numbers of bacteria destroyed by irradiation. If the DEFT count exceeds the APC by 10^4 or more, it indicates that the food has been irradiated (World Health Organisation, 1994). These methods also indicate the hygiene status before and after irradiation, and thus help to enforce Good Manufacturing Practice. They are used for a wide range of irradiated foods, especially to detect irradiated poultry meat. The last method may be limited if initial contamination is low or if low dose levels are used.

Other detection methods are being developed including capillary gel electrophoresis to detect irradiated meat and plant material. Here proteins and peptides are separated on the

basis of their size and this was used to detect changes in egg white caused by irradiation (Day and Brown, 1998).

8.8 Acknowledgement

Grateful acknowledgement is made for information supplied by Isotron plc, Swindon, Wiltshire SN2 6DU, UK.

8.9 References

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